Task 1

1.
```
#include <iostream>
using namespace std;
int main()
{
  char letter;
  letter = 'A';
 Cout << letter << '\n';
  letter = 'B';
 Cout << letter << '\n';
  Cin >> letter;
  return 0;
}
```

output:
```
A
B
```

2 #include <iostream>
using namespace std;
int main()
{
Char letter;
letter = 65;
Cout << letter << '\n';
letter = 66;
Cout << letter << '\n';
Cin >> letter;
return 0;
}

output: A
B
C

3: # include <iostream>
using name space std;
int main ( )
{
int letter;
letter = 'A';
Cout << letter << '\n';
letter 'B';

```
cout << letter << '\n';
cin >> letter;
return 0;
}
```

Output = 65
66

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Coding C++

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RUN MENU

```
{
    char letter;

    // Part 1
    letter = 'A';
    cout << letter << endl;

    letter = 'B';
    cout << letter << endl;

    // Part 2
    letter = 65;
    cout << letter << endl;

    letter = 66;
    cout << letter << endl;

    // Part 3
    int number;

    number = 'A';
    cout << number << endl;

    number = 'B';
    cout << number << endl;

    return 0;
}
```

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Compile Result

```
A
B
A
B
65
66

[Process completed - press Enter]
```

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Task 2

```
#include <iostream>
using namespace std;
int main ( )
int num1, num2;
Cout << Enter two integers: ";
Cin >> num 1 >> num 2;
Cout << "Addition = " << num1 + num2 << endl;
Cout << " Substraction = " << num1 - num2 << endl;
Cout << "multiplication = " << num1 * num2 << endl;
cout << "Division: " << num1/num2 << endl;
    return 0;
}
```

output = Enter two integers : 20 5

Addition = 25

Subtraction = 15

Division = 4

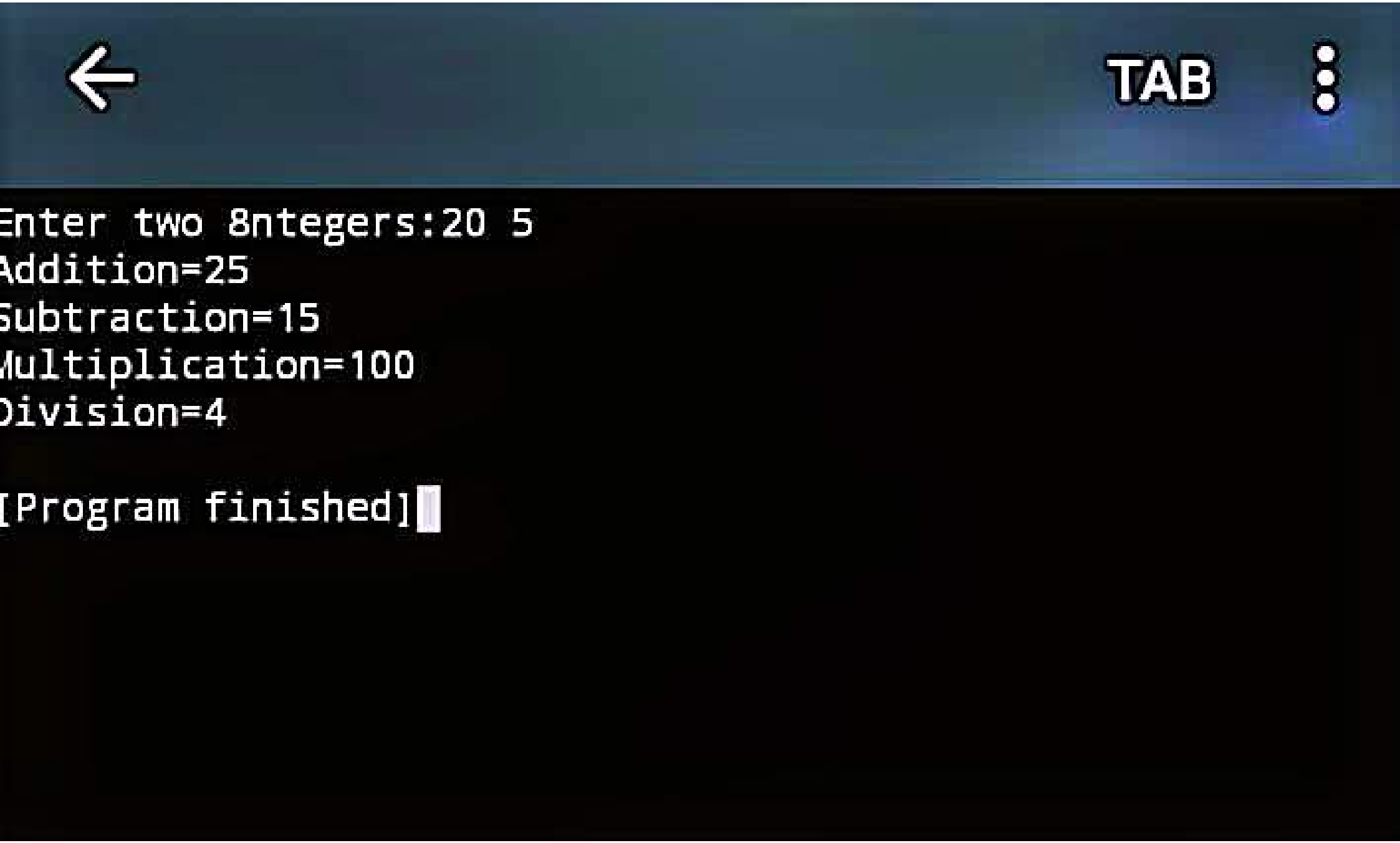
Multiplication = 100

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new*

```
1  # include <iostream>
2  using namespace std;
3  int main ( )
4  {
5  int num1,num2;
6  cout<<"Enter two 8ntegers:";
7  cin>>num1>>num2;
8  cout<<"Addition="<<num1+num2<<endl;
9  cout<<"Subtraction="<<num1-
   num2<<endl;
10 cout<<"Multiplication="<<num1*
   num2<<endl;
11 cout<<"Division="<<num1/num2<<endl;
12 return 0;
13 }
14
```

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TAB
Enter two 8ntegers:20 5
Addition=25
Subtraction=15
Multiplication=100
Division=4
[Program finished]

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Task 3

```
#include "stdafx.h"
#include<iostream>
using name space std;
int main ()
{
 int val1;
 int val2;
 int val 3;
Cout<< please enter your 3 numbers :";
 Cin >> val 1 >>val2 >> val 3;
 Cout << val 1 << "\n" << val 2<<"\n" <val3<<"\n";
 Cout <<val 3<<"\n"<<val 2 <<"\n"<<val 1<<"\n";
   system ("Pause");
     return 0;
   }.
```

output:

```
please enter your 3 num:
10
20
30
30
20
10
```

Task 4

Formula = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

```
#include<iostream>
 # include<cmath>
 using namespace std;
 int main()
{double x1, y1, x2, y2, distance;
 cout << "Enter1 >> y1 ;
Cout <<"Enter x2 and y2: ";
Cin >>x2 >>y2 and y 2:";
distance= Sqrt (Pow(x2-x1 ,2)+ Pow(y2-y1 ,2));
Cout<<"Distance"=<<distance
  return 0;
}
```

output:

```
Enter x1 and y1 : 2 3
Enter x2 and y2 : 5 7
 Distance =5
```

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;

    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    return 0;
}
```

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Compile Result

```
Please enter your 3 numbers: 10 20 30
10
20
30
30
20
10

[Process completed - press Enter]
```

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Coding C++

Auto saved at 02:08:10

RUN MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2;
    double distance;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    distance = sqrt((x2 - x1) * (x2 - x1) +
                    (y2 - y1) * (y2 - y1));

    cout << "Distance = " << distance << endl;

    return 0;
}
```

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Compile Result

```
Enter x1: 2 3 5 7
Enter y1: Enter x2: Enter y2: Distance = 5

[Process completed - press Enter]
```

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TASK 05:

Formula: Area of Rectangle = Length x Width

```
#include <iostream>
using namespace std;
int main ( )
{
 double length, width, area;
 cout << Enter length : ";
 cin >> length;
cout << Enter width : ";
cin >> length;
       width;
area = length * width
cout << Area of Re
= " << area << endl
  return 0;
  }
```

Area of Circle:

Formula: $A = PI \times r^2$

```
# include <iostream>
using namespace std;
int main ( )
{
double r, area;
const double PI = 3.14159;
cout << "Enter radius: ";
cin >> r;
area = PI * r * r;
cout << Area of circle = " << area << endl;
   return 0;
   }
```

Area of Triangle: Area = Base x height x 0.5

```
# include <iostream>
using namespace std;
int main ( )
{
double base, height, area;
cout << Enter base:";
cin >> base;
cout << Enter height: ";
cin >> height;
area = base* height * 0.5;
cout << Area of Triangle = " area << endl;
   return 0;
}
```

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Coding C++

Auto saved at 01:55:58

RUN MENU

```
    circleArea = 3.14159 * radius * radius;

    // Rectangle
    cout << "Enter length of rectangle: ";
    cin >> length;

    cout << "Enter width of rectangle: ";
    cin >> width;

    rectangleArea = length * width;

    // Triangle
    cout << "Enter base of triangle: ";
    cin >> base;

    cout << "Enter height of triangle: ";
    cin >> height;

    triangleArea = 0.5 * base * height;

    // Results
    cout << "\nArea of Circle = " << circleArea << endl;
    cout << "Area of Rectangle = " << rectangleArea << endl;
    cout << "Area of Triangle = " << triangleArea << endl;

    return 0;
}
```

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Compile Result

```
Enter radius of circle: 5 10 4 8 6
Enter length of rectangle: Enter width of r
ectangle: Enter base of triangle: Enter hei
ght of triangle:
Area of Circle = 78.5397
Area of Rectangle = 40
Area of Triangle = 24

[Process completed - press Enter]
```

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TASK 06:

Total purchase

Prices:

Item 1 = $ 12.95
Item 2 = $ 24.95
Item 3 = $ 6.95
Item 4 = $ 14.95
Item 5 = $ 3.95
Sales tax = 6%.

Output:

Item 1: $ 12.95
Item 2: $ 24.95
Item 3: $ 6.95
Item 4: $ 14.95
Item 5: $ 3.95
Subtotal = $ 63.75
Sales tax = $ 3.83
Total: $ 67.58

```
#include <iostream>
#include <iomanip>
using namespace std;
int main()
{
double item1 = 12.95;
double item2 = 24.95;
double item3 = 6.95;
double item4 = 14.95;
double item5 = 3.95;
double subtotal;
double salestax;
double total;
subtotal = item1 + item2 + item3 + item4 + item5
salestax = subtotal * 0.06;
total = subtotal + salestax;
cout << fixed << setprecision (2);
cout "Item 1: $" << item1 << endl;
cout << "Item 2: $" << item2 << endl;
cout << "Item 3: $" << item3 << endl;
cout << "Item 4: $" << item4 << endl;
cout << "Item 5: $" << item5 << endl;
cout << "subtotal: $" << subtotal << endl;
cout << "Sales tax: $" << salestax << endl;
cout << "Total: $" << total << endl;
return 0;
}
```

Coding C++

Auto saved at 01:57:35

RUN MENU

```
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal;
    double salesTax;
    double total;

    subtotal = item1 + item2 + item3 + item4 + item5;
    salesTax = subtotal * 0.06;
    total = subtotal + salesTax;

    cout << fixed << setprecision(2);

    cout << "Item 1 Price = $" << item1 << endl;
    cout << "Item 2 Price = $" << item2 << endl;
    cout << "Item 3 Price = $" << item3 << endl;
    cout << "Item 4 Price = $" << item4 << endl;
    cout << "Item 5 Price = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << salesTax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

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Compile Result

```
Item 1 Price = $12.95
Item 2 Price = $24.95
Item 3 Price = $6.95
Item 4 Price = $14.95
Item 5 Price = $3.95
Subtotal = $63.75
Sales Tax = $3.82
Total = $67.58

[Process completed - press Enter]
```

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BONUS TASK:

Question: Input a five-digit integer and print individual digits with three spaces between them.

```
#include <iostream>
using namespace std;
int main ( )
{
  int number;
cout << Enter a five-digit number: ";
  cin >> number;
cout << number / 10000 << "   ";
cout << (number / 1000) % 10 << "   ";
cout << (number / 100) % 10 << "   ";
cout << (number / 10) % 10 << "   ";
Cout << number % 10 << endl;
   return 0;
}
```

Output:

```
Enter a five-digit number:
12345
1   2   3   4   5
```

Home TASK:

1: Exchange / swap

Question: Read two variables and swap their values.

```
#include <iostream>
using namespace std;
int main ( )
{
 int var1, var2, temp;
cout << Enter value of var1: ";
 cin >> var1;
cout << Enter value of var2: ";
 cin >> var2;
 temp = var1;
 var1 = var2;
 var2 = temp;
cout << After swapping: " << endl;
cout " << "var1 = " << var1 endl;
cout << "var 2" << var2 endl;
cout << "var 3 = " << var3 endl;
 return 0;
}
```

Output:

```
Enter value of var1 = 10
Enter value of var2: 20
After swapping.
var 1 = 20
var2 = 10
```

Coding C++

Auto saved at 02:02:02

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter value of var1: ";
    cin >> var1;

    cout << "Enter value of var2: ";
    cin >> var2;

    cout << "\nBefore swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "\nAfter swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

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Compile Result

```
Enter value of var1: 10 20
Enter value of var2:
Before swapping:
var1 = 10
var2 = 20

After swapping:
var1 = 20
var2 = 10

[Process completed - press Enter]
```

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Coding C++

Auto saved at 02:00:33

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter a five-digit number: ";
    cin >> number;

    int digit1 = number / 10000;
    int digit2 = (number / 1000) % 10;
    int digit3 = (number / 100) % 10;
    int digit4 = (number / 10) % 10;
    int digit5 = number % 10;

    cout << digit1 << "    "
         << digit2 << "    "
         << digit3 << "    "
         << digit4 << "    "
         << digit5 << endl;

    return 0;
}
```

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Compile Result

```
Enter a five-digit number: 12345
1   2   3   4   5

[Process completed - press Enter]
```

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HOME TASK 2: Make Upper

Question:

```
#include <iostream>
using namespace std;
int main ( )
{
char ch;
cout << Enter a lower case alphabet : ";
cin >> ch;
ch = ch - 32;
cout << uppercase alphabet : " << ch << endl;
return 0;
}
```

Output:

Enter a lowercase alphabet : j

Uppercase alphabet : J

Coding C++

RUN MENU

Auto saved at 21:47:59

```cpp
#include<iostream>
using namespace std;

int main()
{
  char letter;
  cout<<"Enter a lowercase alphabet: ";
  cin>>letter;
  letter=letter - 32;
  cout<<"Uppercase alphabet:"
  <<letter<<endl;
  return 0;
  }
```

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Compile Result

```
Enter a lowercase alphabet: b
Uppercase alphabet:B

[Process completed - press Enter]
```

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